

ARNAV DUTTA

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EDUCATION

Master in Science (Hons) in Physics

2023 - 2027

King's College London

Third-Year Modules: Statistical Mechanics, Quantum Mechanics II, Particle Physics, Optics, Condensed Matter Physics I, General Relativity and Cosmology, Advanced Mathematical Methods for Theoretical Physics, Third Year Project in Physics (Modelling Electron Transport in Quantum Materials).

Second-Year Modules: Experimental Physics, Quantum Mechanics I, Electromagnetism, Mathematical Methods for Physics, Introduction to Numerical Modelling, Thermal Physics, Relativity and Sub-Atomic Physics, Mathematical Methods Theoretical Physics.

First-Year Modules: Classical Physics(Electromagnetism, Mechanics and Matter), Mathematics & Computation for Physics, Introduction to Modern Physics, Introduction to Astrophysics, Experimental Physics

A and AS Levels

2021 - 2023

Whitmore High School, London

A-Levels: Further Mathematics, Mathematics, Physics, EPQ (String Theory)

AS Levels: Mathematics, Physics, Chemistry

GCSE

2016 - 2021

Whitmore High School, London

Subjects: Mathematics, English Literature & Language, Triple Science, Computer Science, Geography, Music

TECHNICAL SKILLS

Languages Python (NumPy, SciPy, SymPy, Matplotlib), MATLAB, SQL, Java Script

Software Simulink, LaTeX (Overleaf), Microsoft Office Suite, Fusion 360

Expertise Computational Physics (Monte Carlo Methods, Metropolis Algorithm, Statistical Mechanics), Numerical Methods, Algorithm Design, Data Analysis, CAD Design (Fusion 360), Simulation & Modelling (MATLAB/Simulink & Python)

EXPERIENCE

Tata Technologies

Jul 2025 – Sep 2025

Summer Internship - Warwick, United Kingdom (Hybrid)

- **Contributed** to driver-behaviour analysis and vehicle prognostics projects using real-world car log data.
- **Processed** and **analyzed** large datasets to identify performance patterns and trends.
- **Applied** coding and data analytics techniques to support data-driven insights.
- **Developed** understanding of EV powertrains, battery systems, and diagnostic workflows.
- **Collaborated** with engineers across disciplines, improving communication and project coordination.
- **Automated** directory-level data sorting to streamline large-file organization and reduce manual workload.

KCL Racing, King's College London (Formula Student)

2023 - 2025

Technical Team Member – Powertrain & Steering Team - London, United Kingdom

- **Designed (2024 Car)** drivetrain components, including axle, differential and transmission; contributed to engine systems.
- **Designed (2025 Car)** the steering system, improving handling and integration with the drivetrain.
- **Engineered** cost-effective solutions, optimizing materials and design efficiency within budget constraints.
- **Managed** DBOM, ensuring accurate documentation, meticulous cost tracking, and adherence to a limited budget.

- **Defended** team's Engineering Design and Cost & Manufacturing strategies at Silverstone Formula Student Concept Class (July 2024).
- **Achieved** a milestone as the first-ever King's College London team to compete at Silverstone.

EY Technology Consulting

Aug 2024

Shadowing Placement - London, United Kingdom

- **Shadowed** professionals specializing in Cloud Transformation and AI solutions.
- **Analyzed** best practices in Software Engineering, Cloud Risk, and Control management.
- **Enhanced** knowledge of industry-standard techniques through observation and guided interactions.
- **Gained** insight into real-world technology consultancy processes and workflows.

Tata Consultancy Services (Aerospace & Defence)

Jul 2024

Work Experience Placement - London, United Kingdom

- **Explored** aircraft fundamentals, flight dynamics, and aerodynamic principles.
- **Investigated** mechanisms and functionality of flight control surfaces.
- **Researched** control systems and system-design strategies, enhancing understanding of aviation systems.

Whitmore High School

2021 – 2023

Volunteer Laboratory Assistant - London, United Kingdom

- **Facilitated** GCSE practical sessions in Physics, Chemistry, and Biology.
- **Demonstrated** experiments including titration, photosynthesis, and electric circuits.
- **Prepared** laboratory equipment and materials, ensuring effective hands-on learning in a safe environment.
- **Guided** students individually, clarifying concepts and methods to improve their academic performance.
- **Evaluated** student performance and provided constructive feedback.

UNIVERSITY SOCIETIES

KCL Quantum Mechanics Society, King's College London

2025 - Present

President

- **Leading** the society, overseeing events and initiatives to expand student engagement in quantum physics.
- **Coordinating** committee activities, including problem classes, discussion groups, and book sessions.
- **Organising** academic and social events, balancing learning with community-building opportunities.
- **Fostering** collaboration across year groups and shaping the society's long-term vision while supporting the committee in day-to-day operations.
- **Teaching** core quantum mechanics topics in student sessions, including energy quantisation, the photoelectric effect, angular-momentum quantisation, spin, and spin Hamiltonians.

ACHIEVEMENTS

The Duke of Edinburgh's Award Gold	2024
Residential certificate course in Chalet Cooking at The Orchards School of Cookery	2023
British Physics Olympiad, Experimental Project Silver Award	2020

VOLUNTEERING

The Big Half 2023	2023
Ran the London half marathon (The Big Half 2023), raising funds for Explorers Against Extinction. This event challenged my endurance and allowed me to contribute to conservation efforts.	

HOBBIES

I focus on endurance running, sim-racing, and hiking, with other interests including piano, racket sports, casual gaming, strength training and reading.